

Hazim Al Khusaibi

To: Tommy Devito
Subject: 537 Johnson Avenue_Proposed Retrofit Project

Good Morning Eric/Tommy

Thanks for pursuing us for a possible opportunity to provide you with professional services for the proposed scope of work we discussed. Please see below a summary of all items discussed, a description of the existing property and scope of work proposed to achieve the finished product:

Description of Existing Conditions

The existing premises is located on an entire lot (**Brooklyn Lot #2988**) compromised of the following blocks/buildings:

Lot No.	Address	Size (ft)
1	537 Johnson Avenue	220 x 100
49	557 Johnson Avenue	55 x 135
14	561 Johnson Avenue	180 x 100
42	563 Johnson Avenue	20 x 190
25	218 Randolph Street	45 x 100
37	573 Johnson Avenue	267 x 100

The existing buildings primarily consist of single story 20'-0" high warehouses spaces, a few tenets have installed mezzanines at mid height that are no larger than 25 x 25 feet in size. The boundary load bearing walls are made up of 16" thick masonry walls on all exposures, with common party walls between the above blocks that is a minimum of 12" from - visual inspection.

The roof framing is compromised of roofing (Insulation & 1 ½" Metal Decking) over W14 steel joists that bear on W24 and W30 Girders spanning the entire width in the E-W Direction. The visual inspection of most structural elements indicate that the buildings are in generally good to fair conditions with recent work conducted in some areas.

During the walk through conducted through the premises, a few concerns were raised, primarily sightings of floor slab sagging in some areas and possible cracking in the masonry walls – a thorough inspection of floor slab and wall conditions is highly recommended before performing any of the work described below.

Description of Architectural Proposed Work:

The purpose of the work – as communicated to our office – is to convert the existing warehouses into an indoor (turf soccer) sports facility, with a total of (6) indoor fields on the first floor and (5) fields on the roof. All fields are a standard size of 50 ft x 100 ft for a total of working area of approximately 55,000 square feet. Some of the work includes the removal of load bearing columns and masonry walls in order to create the required open floor plan to fit a grouping of soccer fields. Should the proposed work proceed, it is assumed that architectural plans will be prepared and provided to our office for coordination as work is conducted.

Description of Structural Proposed Work:

Based on our initial site walkthrough, and preliminary analysis – we conclude that the work is possible to perform with the existing structural elements with minimal retrofit (described below). The structural work will focus on (1) Determining the loading capacity of the existing structural elements, (2) Assessing the structural elements for the minimum live load imposed by the fields on the roof, (3) Designing the retrofit and renovation procedures to supplement the existing framing, (4) Designing necessary procedures to introduce or remove load bearing elements to suite field placements.

The itemized list below – *in no particular order* – further describes the necessary work:

1. Field verification of all sizes (CMU wall, deck, steel joists and girders described above)
2. Assess all existing concrete slab on grade for any maintenance required before performing work
3. Assess the roof's capacity to receive a minimum live load of 100 PSF (Based on Gymnasium loads as per NYC Bldg Codes)
4. Determine the adequacy of the items listed below to carry the loads describe above - *compared to the existing roof loads currently imposed*
 - a. Roof Metal Decking

- b. W12 Steel Joists @ 8'-0" O/C
- c. (6) Lines of W24 Girders (Ea. 50 linear feet)
- 5. Depending on the adequacy of the structural elements listed in Step (#3), the following will be prescribed:
 - a. Replacement of the existing roof deck where necessary
 - i. **Worst case scenario: replacement of entire roof deck assembly** (Roof area approx 16,000 sq ft)
 - b. Pour 4 ½" of concrete slab on the installed deck and welded wire mesh
 - c. Install blocking in the form of steel (W12 Beams) between all existing steel joists
 - d. Install beam stiffeners on existing girder web/flange as necessary (*ea. Girder ~50 linear ft*)
 - e. Install diagonal short braces at column-girder connections (*Qty = 12 Columns minimum*)
- 6. Design the removal of the load bearing elements listed below to create required open-space:
 - a. (2) Lines of (5) W10 Columns spaced @ ~30'-0" O/C
 - b. Approx. 60'-0" of Masonry interior walls to open space between 537 and 557 Johnson Avenue
- 7. Design any temporary shoring elements required to provide support while elements listed in Step (#6) are removed or replaced
- 8. Design a new line of columns at the center span of the building to replace the (5) columns removed in Step (#6.a)
 - a. Design required excavation, and foundation of the columns before erection of steel
- 9. Design any retrofit or replacement of girders or girder splice connections as a result of work prescribed in Steps (#3-7)

Summary

Given the information we have so far - the above itemized list of proposed work, is all we could determine at this time, we would reserve the right to recommend any additional work as necessary as we study the property further. If you require any more clarification on the extent of work to be performed, please get in touch and we will be happy to provide more accurate figures. Also review the

Best,

Hazim Al-Khusaibi EIT

Staff Engineer



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